Irfan Habeeb Gazi
Company Name: Google (SWE Intern)
Interview Rounds: One OA Round, followed by two Technical Interview Rounds

Interview Process:
OA Round: The OA consisted of 2 DSA questions and was conducted on HackerRank. The first question was based on dynamic programming on trees using sieve. The second question was an array question where you had to find the number of partitions in the array meeting some mentioned conditions, the question can be solved using maps.
I had solved the first question but wasted too much time on it. I was unable to write the optimal solution for the second question (TLE).

Technical Interview 1: The interviewer was conducted on Google Meet and we had to use Interview Docs, which is similar to Google Docs but with syntax highlighting for code.
The interviewer introduced herself and then asked for my introduction.
Then we moved to the DSA questions. The first question was fairly straightforward and was a variation of the Jump Game question on LeetCode. I was able to come-up with the solution and code it in 15 minutes. The interviewer then asked me a follow up question, where some additional restrictions were placed on what moves I can make. I solved this using backtracking. The interviewer seemed happy with my solution.
Finally, I was asked if I had any more questions and I asked her a bit about the internship process at Google and the projects we would get to work on.

Technical Interview 2: The interviewer directly jumped into the question in this one. The first question was similar to the Rain Water Trapping problem(1D array). But instead of rain there were fountains on some select buildings and we had to find which buildings were flooded. The question provided at start was a bit obscure and you were required to probe a bit to understand the complete question. I found an optimal solution using monotonic stack and difference array and coded up the solution. The interviewer then had a follow up question on this where we were given a 2D array of houses. I was stuck on this for a while, luckily the interviewer hinted me towards using graphs, and I was able to come up with the optimal solution using DFS. Unfortunately, we were short of time by then, so I was just asked to dry run my DFS solution. Overall, the interviewer seemed satisfied with the approach.

VERDICT: Selected

Preparation Tips: I strongly recommend being thorough with all basic DSA algorithms including Graphs and D inynamic Programming. Striver’s A2Z Sheet and practicing on LeetCode should be sufficient for this. For all the questions in the Interview I was asked to dry run my code, so it is important to be comfortable with it.

Soumyadipto Pal
Company Name: Google(SWE Intern)
Rounds: Online Assessment, followed by 2 Technical Interview Rounds

Online Assessment:

The online assessment included two DSA questions, and I had 90 minutes to solve them. The first question was a medium-level problem where I had to optimize the cost of delivering goods to ‘n’ places based on some given conditions. This could be efficiently solved using two priority queues. I managed to get the correct solution for this one.The second question involved arrays and required finding the number of partitions in the array that meet certain conditions. The optimal solution involved using maps, but I couldn't figure it out in time, so I wrote a brute-force approach instead. This only got me partial credit.

Interview Round 1:

Both interview rounds were conducted over Google Meet, and we used a shared document for coding. My interviewer started by introducing herself, and then asked me to introduce myself.The first question was pretty straightforward: I was given a graph and asked to select a vertex that, when assigned as the root, would make the graph a binary tree.The second question was a follow-up to the first. This time, the edges of the graph were colored black or white, and I needed to select a root edge that would create a binary tree with alternating colors at each level (e.g., Level 1 is black, Level 2 is white, and so on).I was asked for the most optimized solution(in O(N) time). After solving both problems, I was asked to explain the space and time complexity of my solutions.

Interview Round 2:

In the second interview round, I was asked to sort an array based on certain conditions, but I had to do it in less than O(nlogn) time. This could be accomplished using a priority queue. The second question was about a tree where each edge had a value of either zero or one, representing water and land, respectively. I needed to find out the number of islands. For both questions, the interviewer asked me to do a dry run and explain the time and space complexity. Both problems were relatively easy, and the interview wrapped up in about 30 minutes.

Verdict: Selected

Preparation Tips: Practice all the basic DSA algorithms on graphs and trees. Learn the space and time complexity of the algorithms. Practice dry running your solutions. During the interview, interact with the interviewer and ask as many questions as possible. Think aloud while solving the problems, and try to explain your code as you write.

Brahmajit Ray
Company Name: Google(SWE Intern)
Rounds: Online Assessment, followed by 2 Technical Interview Rounds

Online Assessment:

The online assessment was of 90 minutes and had 2 DSA Questions.The first question was pretty close to LeetCode 1383(Maximum Performance of a Team) but was written in a confusing manner,so it took me some time to be able to map that problem to the problem that I had solved before.The second question was a question on strings and needed to find some partitions that met certain conditions.I was partially able to solve it with backtracking and ended up with a TLE.

Round 1:

All Interviews for Google were conducted over Google Meet.In the presentation we had been told that we would be assessed on only and only our DSA Skills,so that streamlined the preparation for the interview. With the interview link,a link to an Interview Doc was shared,which is in appearance very similar to the Python IDLE(has nothing other than syntax highlighting). The interviewer jumped straight to the question and said there is no point wasting 5-6 min in a high pressure interview introducing ourselves(I really liked this as Google is really stringent about the interview window).The question asked was a involved a 2D matrix over which a sliding window approach with prefix sum had to be used.I wrote the code and was asked to dry run it.The interviewer was happy with the approach.

Round 2:

Ok,this is where the screw ups began for me.First of all, after my first round interview at around 4pm that day, I had not received any links or confirmation about my 2nd round interview.However at 3:30am(I luckily was up),I received a mail with an interview link and it was scheduled at 10am the next morning.The next morning,during the interview,the doc wouldn't open,neither for me nor the interviewer.Then,I probably faced what I would say to be the worst question on strings I have ever seen.For about 20 minutes, I absolutely had no clue about how to approach the problem. The interviewer himself acknowledged that it was a pretty tough problem and that he would give me extra time for that doc mishap and the overall difficulty of the problem. He started guiding me through the problem and then I somehow managed to drag myself to a $O(n^3)$ solution that I was able to further optimize to $O(n^2 \log n)$.I ran out of time and the interviewer then told me that the best solution was $O(n)$ but acknowledged that he himself did not expect it from a candidate, as it required a rather uncommon algorithm and was not at all intuitive and that he would be fine with $O(n^2)$.

Verdict: Not Selected